

History $\mathcal{H}_n$

every evaluated particle $i \leq n$

Archive A_n
the k best by ρ

never reported

Adaptation weights

$\tilde{W}_j^{(n)} \propto w_j K_{\epsilon_n}(\rho_j), \quad j \in A_n$
(w_j : the stored proposal-time weight)
→ parent selection, covariance Σ_n
→ the next proposal q_n

never enters the posterior

Proposal-time weight

$w^\star = \pi(\theta^\star) / \bar{q}_n^{\text{on}}(\theta^\star)$
(balance heuristic over the $S=20$ snapshot buffer)
→ stored with the particle as w_i
→ online ESS diagnostic

the only weighting reported

Reported weights (retroactive, over the whole history)

$$W_{i,n} = \frac{\pi(\theta_i) K_{\epsilon_n}(\rho_i)}{\bar{q}_n(\theta_i)}, \quad i \leq n$$

$\bar{q}_n = m$ history-reconstructed proposals + prior floor
recomputed from $\mathcal{H}_n$ on demand, $O(nk)$ once

next proposal
 q_n

draw

candidate $\theta^\star$
evaluated → $\rho^\star$

$\hat{\pi}_n(f)$
reported posterior

kernel weights over A_n

appended to the log

all n particles